Canonical Components and Generalized Spherical Harmonics

Conventions, symmetries and admissible operations

Notes for the GSHTrans field algebra

Abstract

This note fixes the conventions used by the GSHTrans field algebra and derives the facts that the type system is intended to enforce: how the generalized spherical harmonic upper index is carried by a tensor component, which algebraic operations on components are covariant, and how reality and permutation symmetry reduce the set of components that must be stored. Every convention-dependent statement is flagged; these are the points that need checking against Phinney & Burridge before any code is written.

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1 The canonical basis

At each point of the sphere let $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$ be the usual orthonormal spherical polar frame. The canonical basis is

$$\mathbf{e}_\pm = \mp\tfrac{1}{\sqrt{2}}\left(\hat{\boldsymbol{\theta}} \pm i\hat{\boldsymbol{\phi}}\right), \qquad \mathbf{e}_0 = \hat{\mathbf{r}}. \tag{1}$$

The index $\alpha \in \{-1, 0, +1\}$, written $\{-, 0, +\}$ where no confusion arises.

Convention check. The sign $\mp$ in (1) is the first thing to verify. Some authors use $\mathbf{e}_\pm = \pm(\hat{\boldsymbol{\theta}} \pm i\hat{\boldsymbol{\phi}})/\sqrt{2}$, which flips the sign of every odd-rank reality phase below.

Two properties follow immediately and are used throughout. First, complex conjugation of the basis:

$$\overline{\mathbf{e}_\alpha} = (-1)^\alpha\, \mathbf{e}_{-\alpha}, \tag{2}$$

which reproduces $\overline{\mathbf{e}_\pm} = -\mathbf{e}_\mp$ and $\overline{\mathbf{e}_0} = \mathbf{e}_0$. Second, the metric $g_{\alpha\beta} = \mathbf{e}_\alpha \cdot \mathbf{e}_\beta$:

$$g_{\alpha\beta} = (-1)^\alpha\, \delta_{\alpha+\beta,\,0}, \qquad \text{that is} \qquad g_{+-} = g_{-+} = -1, \quad g_{00} = 1, \tag{3}$$

all other entries vanishing. Note g is its own inverse. We work throughout with *contravariant* components only; the metric (3) is applied explicitly wherever a contraction is taken, and index position is not tracked.

2 Canonical components of a tensor

A rank-p tensor field on the sphere is expanded as

$$\mathbf{T}(\theta, \phi) = \sum_{\alpha_1 \dots \alpha_p} T^{\alpha_1 \dots \alpha_p}(\theta, \phi)\, \mathbf{e}_{\alpha_1} \otimes \cdots \otimes \mathbf{e}_{\alpha_p}, \tag{4}$$

so a rank-p tensor has 3^p scalar components, each labelled by a *multi-index* $(\alpha_1 \dots \alpha_p) \in \{-1, 0, 1\}^p$.

Under a rotation of the tangent frame through an angle ψ about $\hat{\mathbf{r}}$ the basis vectors transform as $\mathbf{e}_\pm \mapsto e^{\mp i\psi}\mathbf{e}_\pm$, $\mathbf{e}_0 \mapsto \mathbf{e}_0$. Hence the component $T^{\alpha_1 \cdots \alpha_p}$ acquires the phase $e^{-iN\psi}$ with

$$\boxed{N = \sum_{i=1}^{p} \alpha_i} \tag{5}$$

and it is this N, not the multi-index itself, that is the upper index of the generalized spherical harmonic expansion of that component (Section 4).

Two consequences matter for the design of the library.

1. For a vector ($p = 1$) the multi-index and the upper index coincide, $N = \alpha$. For $p \geq 2$ they do not: distinct components share an upper index. A rank-2 tensor has three components with $N = 0$, namely $(-+)$, (00) and $(+-)$. A collection of components labelled only by N therefore does *not* determine the tensor.
2. The number of components of a rank-p tensor with a given upper index N is the trinomial coefficient $\binom{p}{N}_2$; for $p = 2$ these are $1, 2, 3, 2, 1$ for $N = -2, \dots, +2$.

N	components	count
-2	$(--)$	1
-1	$(-0),\ (0-)$	2
0	$(-+),\ (00),\ (+-)$	3
$+1$	$(0+),\ (+0)$	2
$+2$	$(++)$	1

Table 1: Upper index of each component of a rank-2 tensor.

Tensor product. If S has multi-index $(\alpha_1 \dots \alpha_p)$ with upper index N_S and T has $(\beta_1 \dots \beta_q)$ with N_T, then the component of $S \otimes T$ with the concatenated multi-index has upper index $N_S + N_T$ by (5). The rule "upper indices add under pointwise multiplication" is exactly this statement.

3 Reality

Suppose $\mathbf{T}$ is a real tensor field. Writing $\mathbf{T} = \overline{\mathbf{T}}$ and using (2) on each factor,

$$\sum_{\alpha_1 \dots \alpha_p} T^{\alpha_1 \dots \alpha_p}\, \mathbf{e}_{\alpha_1} \otimes \cdots = \sum_{\alpha_1 \dots \alpha_p} \overline{T^{\alpha_1 \dots \alpha_p}}\, (-1)^{\alpha_1 + \cdots + \alpha_p}\, \mathbf{e}_{-\alpha_1} \otimes \cdots, \tag{6}$$

and relabelling $\alpha_i \to -\alpha_i$ on the right gives the **reality condition**

$$\boxed{T^{-\alpha_1\ldots-\alpha_p} = (-1)^N\,\overline{T^{\alpha_1\ldots\alpha_p}}, \qquad N = \textstyle\sum_i \alpha_i.} \tag{7}$$

For a vector this reads $u^0 = \overline{u^0}$ and $u^- = -\,\overline{u^+}$: the radial component is a real field, and the two transverse components are not independent.

3.1 Which components must be stored

Equation (7) pairs each multi-index with its *negation* $(\alpha_i) \mapsto (-\alpha_i)$. The components that must be stored are one representative of each orbit of this involution.

- The only fixed point of the negation map is the all-zero multi-index, which has $N = 0$ and, by (7), satisfies $T^{0\ldots0} = \overline{T^{0\ldots0}}$: it is a *real-valued* field.
- Every other multi-index lies in an orbit of size two; one member is stored as a complex field and the other is recovered by (7).

The storage cost in real numbers is therefore $1 + 2\cdot\frac{1}{2}(3^p - 1) = 3^p$, which is precisely the number of real degrees of freedom of a real rank-p tensor in three dimensions. The reduction loses nothing and saves a factor of two against storing all 3^p components as complex fields.

rank p	components 3^p	stored fields	of which real	reals stored
0	1	1	1	1
1	3	2	1	3
2	9	5	1	9
4	81	41	1	81

Table 2: Storage for a real tensor field, per grid point, under the orbit rule of Section 3.1. Complex tensors of the same rank require $2\cdot 3^p$ reals.

Why "$N \geq 0$" is not the rule. For a vector the orbits are $\{0\}$ and $\{+,-\}$, and selecting components with $N \geq 0$ picks exactly one from each. This coincidence fails at rank 2: $(+-)$ and $(-+)$ both have $N = 0$ and form a negation pair, so the criterion $N \geq 0$ would retain both. The selection rule must be stated on multi-indices, not on upper indices.

3.2 Interaction with permutation symmetry

If the tensor is symmetric (or antisymmetric) under some subgroup P of index permutations, the stored set is one representative per orbit of the group generated by P together with the negation map. A component whose P-orbit contains its own negation is constrained by (7) to be real, or purely imaginary if the permutation relating them carries a minus sign.

Such self-paired components always have $N = 0$: permutation preserves the multiset sum while negation changes its sign, so a component fixed by the combined operation satisfies $N = -N$. Since $(-1)^0 = +1$, a self-paired component of a *symmetric* tensor is always real; purely imaginary components arise only in the antisymmetric case.

Worked check: symmetric real rank 2. The six independent symmetric components are $(--), (-0), (-+), (00), (0+), (++)$. Under the combined group,

orbit	size	stored as
$\{(00)\}$	1	real field
$\{(-+),(+-)\}$	2	real field (self-paired, $N=0$)
$\{(--),(++)\}$	2	complex field
$\{(-0),(0-),(+0),(0+)\}$	4	complex field

giving $1+1+2+2=6$ real numbers per point, which is the number of independent entries of a real symmetric 3×3 matrix. That the orbit construction reproduces this without special-casing is a useful check that the abstraction is the right one.

4 Generalized spherical harmonics

A field f of upper index N is expanded as

$$f(\theta,\phi)=\sum_{l\geq|N|}\sum_{m=-l}^{l} f^N_{lm}\,Y^N_{lm}(\theta,\phi),\qquad f^N_{lm}=\int_{S^2}\overline{Y^N_{lm}}\,f\,\mathrm{d}\Omega, \tag{8}$$

with

$$Y^N_{lm}(\theta,\phi)=\sqrt{\frac{2l+1}{4\pi}}\;d^l_{mN}(\theta)\,e^{im\phi} \tag{9}$$

in the `Ortho` normalisation, where d^l_{mN} is the real Wigner d-function. These are orthonormal on the sphere, $\int\overline{Y^N_{lm}}Y^N_{l'm'}\,\mathrm{d}\Omega=\delta_{ll'}\delta_{mm'}$. The sum over l starts at $|N|$ because d^l_{mN} vanishes identically for $l<|N|$.

Convention check. The index order in d^l_{mN}, and whether the code's `Wigner` class stores d^l_{mn} or d^l_{nm}, should be confirmed against `src/Wigner.h`. The normalisation in (9) is however pinned by the $l=0$ behaviour of the existing transform: taking f constant, (8) gives $f^0_{00}=2\sqrt{\pi}f$ and $f=f^0_{00}/(2\sqrt{\pi})$, which is what `GaussLegendreGrid` implements on its one-point grid.

4.1 Reality in the spectral domain

Using $d^l_{-m,-N}=(-1)^{m-N}d^l_{mN}$ and the reality of d^l_{mN}, equation (9) gives

$$\overline{Y^N_{lm}}=(-1)^{m+N}\,Y^{-N}_{l,-m}. \tag{10}$$

Two distinct consequences follow, and it is worth keeping them apart.

Basis-level. For any field f of upper index N, the coefficients of $\overline{f}$, which has upper index $-N$, are

$$\left(\overline{f}\right)^{-N}_{l,-m}=(-1)^{m-N}\,\overline{f^N_{lm}}. \tag{11}$$

This is the relation exercised by `TestRealFieldSymmetry`.

Component-level. For a component pair of a *real* tensor, (7) supplies an extra factor $(-1)^N$, and the two combine to give

$$\boxed{T^{-N}_{l,-m}=(-1)^m\,\overline{T^N_{lm}}} \tag{12}$$

for the coefficients of the negation-paired components. At $N=0$ this reduces to the familiar condition $f_{l,-m}=(-1)^m\overline{f_{lm}}$ for a real scalar field, whose reduced $m\geq 0$ storage the library already implements.

5 Admissible algebraic operations

An operation on canonical components is admissible only if its result is again a component of definite upper index, that is, only if it commutes with the frame rotation $\mathbf{e}_\pm \mapsto e^{\mp i\psi}\mathbf{e}_\pm$. Writing $[f] = N$ for the upper index of f:

operation	upper index of result	restriction
$f+g$, $f-g$	N	requires $[f] = [g] = N$
fg	$[f] + [g]$	—
f/g	$[f]$	requires $[g] = 0$
sf, $s \in \mathbb{C}$	$[f]$	—
$\overline{f}$	$-[f]$	—
$\lvert f\rvert$, $\lvert f\rvert^2$	0	—
$\operatorname{Re} f$, $\operatorname{Im} f$	0	requires $[f] = 0$
$F(f)$, F nonlinear	0	requires $[f] = 0$
$\int_{S^2} f \,\mathrm{d}\Omega$	—	requires $[f] = 0$

Three of these deserve comment, because the present implementation violates them.

- **Conjugation reverses the upper index.** A quantity carrying $e^{-iN\psi}$ has a conjugate carrying $e^{+iN\psi}$. This is consistent with (7): u^-, of upper index -1, is $-\overline{u^+}$ and u^+ has upper index $+1$.
- **Real and imaginary parts are defined only at $N = 0$.** For $N \neq 0$ the split of f into real and imaginary parts is not preserved by a frame rotation, so neither part is a component of any tensor. What *is* available at any N is the modulus, which is a genuine scalar.
- **Nonlinear functions likewise require $N = 0$.** Only for $N = 0$, where the component is invariant, is $F(f)$ again a component. Integration over the sphere is subject to the same restriction, since $\int f \,\mathrm{d}\Omega$ vanishes identically unless $N = 0$.

Contraction. Contracting the jth and kth slots of a rank-p tensor uses the metric (3):

$$(\mathrm{tr}_{jk}\, T)^{\cdots} = \sum_{\alpha\beta} g_{\alpha\beta}\, T^{\cdots\,\alpha\,\cdots\,\beta\,\cdots} = \sum_{\alpha} (-1)^\alpha\, T^{\cdots\,\alpha\,\cdots\,-\alpha\,\cdots}, \tag{13}$$

the contracted pair contributing $\alpha + (-\alpha) = 0$ to the upper index, so the result has the upper index of the surviving slots, as it must.

6 Raising and lowering operators

These are not required for the first stage of the field algebra but are recorded here because they constrain its design: they are the operations that connect different upper indices, and they are diagonal in (l, m) rather than local in (θ, ϕ).

The operators $\eth$ and $\bar{\eth}$ raise and lower the upper index by one. Acting on a basis function of upper index N,

$$\eth\, Y^N_{lm} = -\sqrt{(l-N)(l+N+1)}\; Y^{N+1}_{lm}, \qquad \bar{\eth}\, Y^N_{lm} = +\sqrt{(l+N)(l-N+1)}\; Y^{N-1}_{lm}, \tag{14}$$

so that in the spectral domain each is a multiplication of the coefficients by an l-dependent factor, with no coupling between different (l, m).

Convention check. The overall signs in (14), and the relation between Y^N_{lm} and the spin-weighted harmonics ${}_sY_{lm}$ (whether $s = N$ or $s = -N$), vary between sources and must be fixed against Phinney & Burridge. The magnitudes are not in doubt.

The surface gradient of a scalar field f has canonical components proportional to $\eth f$ and $\bar{\eth} f$, of upper index $+1$ and -1 respectively, with vanishing radial component; surface divergence and curl of a vector field are the corresponding contractions. The practical consequence for the library is that an expression such as "the gradient of a product" is necessarily evaluated in two representations: the product is local in (θ, ϕ), the gradient is local in (l, m).

7 Summary of facts the type system should enforce

1. A component is labelled by a multi-index; its upper index is the signed sum (5) and is a compile-time quantity.
2. Addition requires equal upper indices; multiplication adds them; division requires a denominator of upper index zero.
3. Conjugation maps $N \mapsto -N$.
4. Real part, imaginary part, nonlinear functions and integration over the sphere require $N = 0$.
5. A real-valued component field can occur only at $N = 0$.
6. For a real tensor, the stored components are one per orbit of index negation, combined with any permutation symmetry; the all-zero component is real-valued and the remainder are complex.